

Differential returns to securitization: Evidence and impact

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Where do mortgage lenders' rents come from without deposits?

- Nonbanks originate **two-thirds** of U.S. mortgages (over 90% of FHA/VA), and balance-sheet lending is very small
 - Despite this, the IO/banking literature treats the “Originate-to-Distribute” (OTD) supply chain as a “black box”
- Mortgage supply chain: buy/originate loans, hold for weeks, resell as a security
⇒ OTD margin = **selling high** (differentiated securities) – **buying low** (loan acquisition)
- We study the sources of market power that arise from producing and delivering **differentiated** securities

Ginnie Mae MBS: a clean setting to study returns to securitization

- Ginnie Mae issuers, banks and nonbanks backing FHA/VA loans, choose between **two securitization channels**:
 - **TBA market** (75% of loans): large multi-issuer pools, liquid, but priced on *coarse* characteristics only (coupon, maturity, agency)
 - **Spec-pool market** (25% of loans): single-issuer pools, *differentiated* by loan size — within a segment, investors pay a premium for pools of longer-duration loans

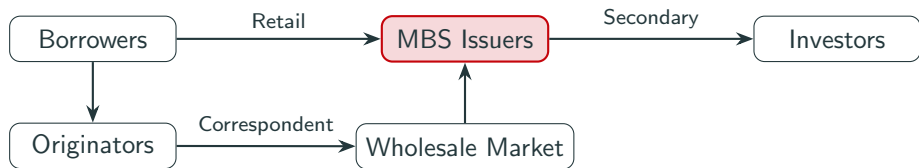
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- **Customization**: Place longer-duration loans in spec-pools, short-duration in TBA
 - ⇒ Spec-pool prices reflect loan quality — issuers earn a **pay-up**
 - ⇒ Example: Loans < 100K vs > 200K in April 2019
 - 12-month survival probability is **50%** larger for smaller loans
 - Revenue from securitization: **\$6/100 vs \$3/100**

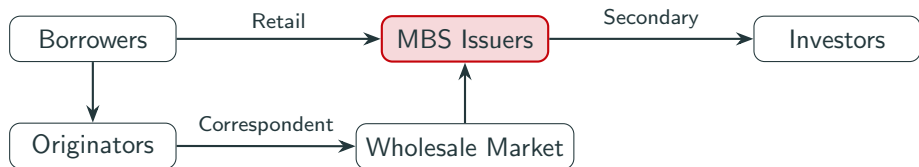
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 - Revenue from securitization: **\$6/100 vs \$3/100**
- **Important fact**: Median issuer securitizes 1.5% of its portfolio in spec-pools, and returns from securitization are very dispersed across issuers

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- **Research question:** is the pay-up passed through to originators and borrowers?
- *If retail/wholesale markets are competitive:* yes — a transfer from large-loan to small-loan borrowers
- **Our finding:** no — customization creates market power rents

Mechanism generalizes: Freddie/Fannie segment, CMBS, auto loans, credit cards

Contribution and findings

What do we do?

- Test predictions from a model of issuer customization for Ginnie Mae securities
- Leverage first data set linking: retail → wholesale → securitization → performance

Key findings:

- **Heterogeneous** gains from customization due to scale and portfolio duration
- Asymmetries between MBS issuers generate **market power** — pass-through of 44% for wholesale, near zero for retail
- **Counterfactual:** eliminating customization — large issuers lose (-\$172/loan), small issuers and borrowers gain

Policy implications for:

- MBS **market design** (e.g. Uniform MBS reform) and **nonbank regulation**

Outline

Institutions and data

Loan valuation model

Empirical analysis:

- Adverse selection

- Differential returns

- Pass-through analysis

Counterfactual: eliminating adverse selection of multi-issuer pools

Policy implications

Data: linking prices across the supply chain

Market	Price	Data Source	Sample
Retail Borrower $\rightarrow$ Originator	$P_i(r) = 100 - \text{Net Fee}(r)$	HMDA (2018–2019)	2M loans 60K auctioned loans
Wholesale Originator $\rightarrow$ Issuer	$W_i = \Delta^W + V_i(c)$	Optimal Blue (OB) (2018–2019)	70K auctions (1st-price sealed bid)
Secondary Issuer $\rightarrow$ Investors	$V_{t(i)}^{\text{tba}}(c)$ $V_i^{\text{sp}}(c, \bar{Z})$	Bloomberg TRACE + eMBS	60K securities 14M loans (2013–22)

Key definitions:

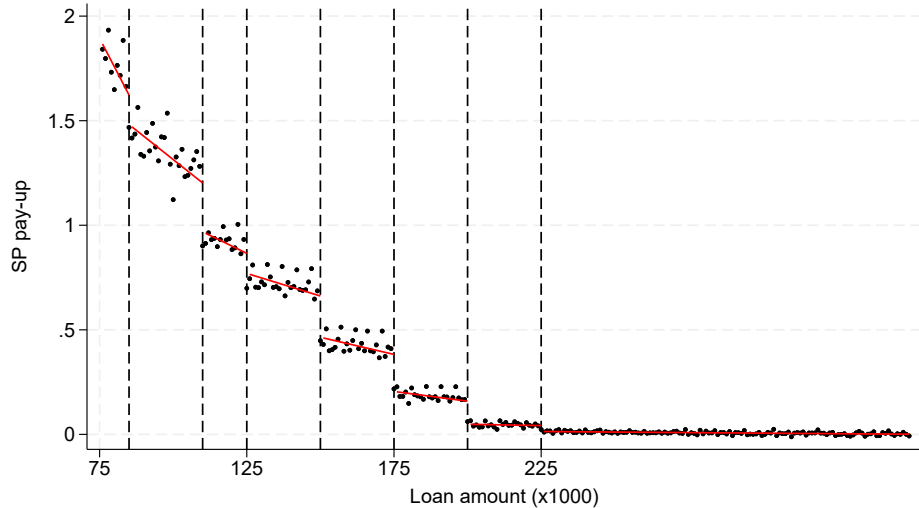
- Δ^W = Wholesale pay-up (issuer markup over V_i)
- $\Delta^{SP} = V_i^{\text{sp}}(c, \bar{Z}) - V_{t(i)}^{\text{tba}}(c)$ = Spec-pool pay-up (return to customization)
- **Resulting data set:** retail $\rightarrow$ wholesale $\rightarrow$ securitization $\rightarrow$ performance

Fact 1: Prepayment risk and customization vary across loan sizes

Size	Share	% TBA	Survival (%)	
			12mo	36mo
$\leq 85K$	4.4	0.210	0.950	0.584
85–110K	4.6	0.205	0.886	0.406
110–125K	4.4	0.285	0.851	0.386
125–225K	40.0	0.666	0.783	0.250
$> 225K$	46.6	0.961	0.619	0.115
Total	100.0	0.745	0.722	0.215

Sample: All Ginnie Mae loans securitized in April 2019.

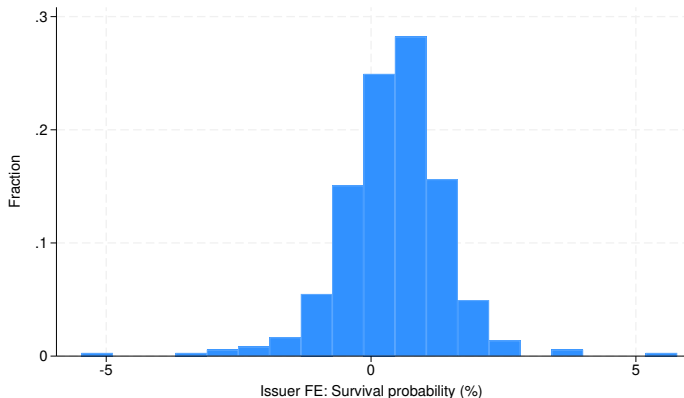
Fact 2: segmentation by loan size creates discontinuous spec-pool pay-ups



Bandwidth = \$1,000

Fact 3: issuers deliver loans with systematically different durations

$$100 \times 1(T_i > 12) = Z_i\beta + \text{Issuer FE} + \text{Rate} \times \text{Date FE} + \text{Error}$$



Magnitude: The 10–90 spread in issuer effects is as large as the loan-size gradient from \$85K to \$225K, and about **twice** the effect of a quarter-point (25 bp) increase in the note rate. [▶ Lender characts](#)

Model of loan valuations

- **Setup:** Two-stage decision
 1. Acquisition of loans in the wholesale and/or retail market
 2. Customization decision: Spec or TBA pool
- Willingness to pay (WTP) from acquiring a \$100 loan:

$$WTP = \underbrace{(r - c - g) \times \text{Loan duration } (\bar{M})}_{\text{Mortgage servicing rights (MSR)}} + \underbrace{\text{Resale value}}_V - \underbrace{\text{Cost} + \text{Retention option}}_{\text{Net fixed-cost } (F)}$$

- Important: Focus on loans with no coupon choice (75%)

- **Assumption:** Competitive spec-pool prices:

$$V(c, q) \equiv V(c, Z_1, \dots, Z_q) = \frac{1}{\sum_{i=1}^q L_i} \sum_{i=1}^q L_i v_c^{(i)}, \quad \text{with } \partial V(c, q) / \partial q < 0$$

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- Optimization problem for loans in a given size segment, e.g. $L \in [125 - 150]$

$$\max_{q \in [0, Q_c]} qV(c, q) + (Q_c - q)V^{\text{tba}}(c) - \kappa 1(q > 0)$$

- $q^* \in (0, Q_c)$: Marginal loan $v_c^{q^*} = V^{\text{tba}}(c)$
- $q^* = Q_c$: Worst loan $v_c^{Q_c} > V^{\text{tba}}(c)$
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- Why do firms have different customization strategies?
 - $\Rightarrow$ Economies of scale and portfolio duration (i.e. quality)

- **Implication:** Heterogeneity in scale and duration leads to differences in WTP
- **Wholesale market power:**
 - *Small loans:* $\uparrow$ markup and *incomplete* pass-through of security prices
 - *Large loans:* $\downarrow$ markup and *complete* pass-through of security prices
- **Retail market power:**
 - Information frictions **and** originator heterogeneity $\uparrow$ markup and $\downarrow$ pass-through of security prices

Prediction 1: selection into spec-pools depends on issuer portfolio duration

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Under our competitive secondary market assumption, the model predicts:

1. Issuers select into spec-pools based on ALL observable attributes (not just size)
2. Selection reflects issuer-specific portfolio duration

Test: Chiappori & Salanié (2000), Finkelstein & Poterba (2004, 2014)

- Correlation between realized duration and selection into multi-issuer pools:

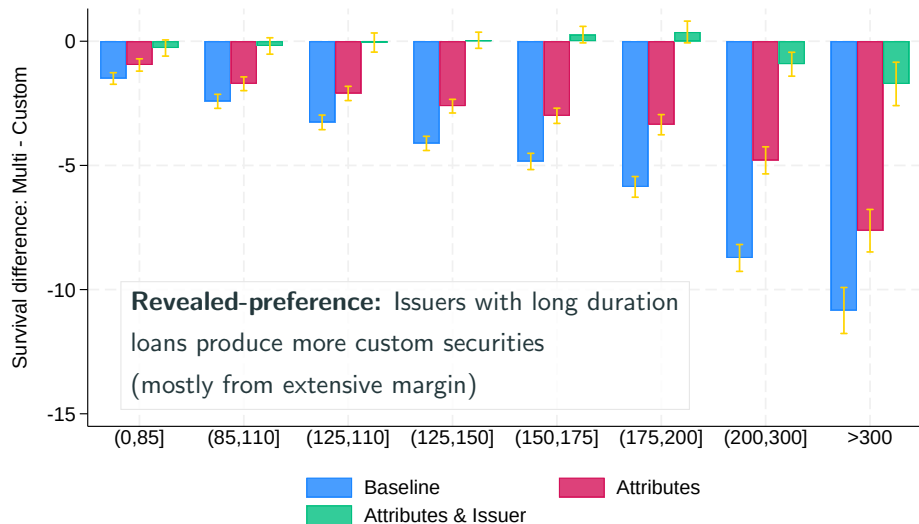
$$100 \times 1(T_i > 12) = \lambda 1\{\text{Multi-issuer MBS}\} + \beta Z_i + \text{Rate} \times \text{Date FE} + \epsilon_i$$

where T_i is realized duration and Rate $\times$ Date FE captures information available in the TBA market.

Key insight: By varying Z_i , we can decompose the selection mechanism

- Loan attributes VS Issuer heterogeneity VS Private information

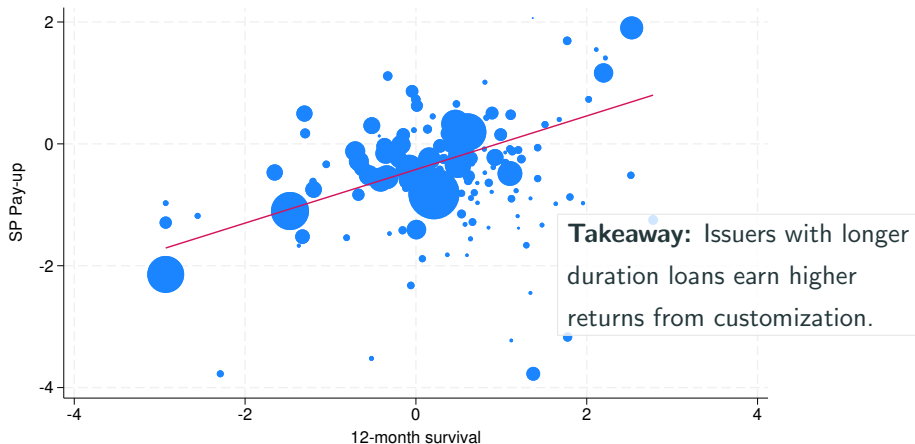
Decomposing the selection mechanism



Prediction 2: issuer scale and duration explain differences in returns from securitization

- **Model predicts:** Customization rates and returns vary with firm size and portfolio duration
- **Step 1:** Extract issuer fixed effects to measure firm heterogeneity (Chandra et al.2016)
- **Step 2:** Correlate customization FEs with firm characteristics

Prediction 2a: security prices vary with portfolio duration



Variance decomposition: Issuer heterogeneity and loan characteristics are roughly equally important at explaining SP prices (13% vs 15%). [► Table](#)

Prediction 2b: economies of scale in customization

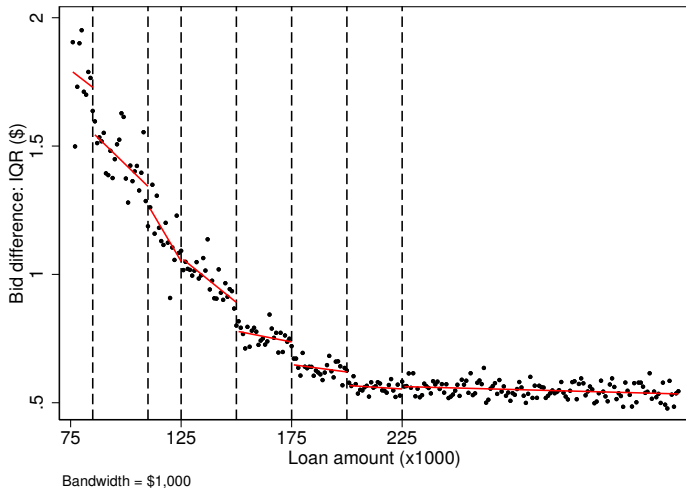
VARIABLES	(1) Customization	(2) Customization	(3) Customization	(4) Customization
SP Pay-up FE				0.159* (0.0854)
Monthly volume (log)	0.288* (0.162)	0.413*** (0.0658)	0.311*** (0.0643)	0.316*** (0.0690)
Survival FE		0.385*** (0.0566)	0.388*** (0.0421)	0.309*** (0.0468)
Retail origination FE			-0.154*** (0.0553)	-0.172*** (0.0561)
Constant	-1.246** (0.596)	-1.726*** (0.260)	-1.300*** (0.270)	-1.268*** (0.280)
Observations	133	133	133	133
R-squared	0.090	0.534	0.582	0.598

Observations are weighted by the number of loans per issuer. SE clustered by issuer. * $p < .1$; ** $p < .05$; *** $p < .01$. Weighted version. All variables standardized.

Prediction 3: incomplete pass-through in wholesale and retail

- **Question:** Does heterogeneity in returns from customization translate into market power?
- **Approach:**
 1. Unobserved heterogeneity in WTP: Within auction dispersion in bids
 2. Retail and wholesale pass-through regressions

Heterogeneity in WTP: bids are more dispersed in auctions for small loans



Sample: Optimal-Blue wholesale auctions from 2018–2019.

► Difference: 1st - 2nd bids

Pass-through: How are returns from customization distributed?

- Linear pass-through model:

$$W_i = f(Z_i) + \mu_{\text{pay-up}}^w \left[V_i - V_t^{\text{tba}}(c) \right] + \mu_{\text{tba}}^w V_t^{\text{tba}}(c) + \text{FE} + \text{Error}$$

$$P_i = g(Z_i) + \mu_{\text{pay-up}}^r \left[W_i - V_t^{\text{tba}}(c) \right] + \mu_{\text{tba}}^r V_t^{\text{tba}}(c) + \text{FE} + \text{Error}$$

Where:

- *Security price*: $V_i = V_{t(i)}^{\text{tba}}(c)$ [Multi-issuer] or $V_i^{\text{sp}}(c)$ [Custom]
- *Wholesale price*: $W_i = \text{Highest bid}$
- *Retail price*: $P_i = 100 - \text{Lender fee} + \text{Discounts}$
- **Note**: $V_i^{\text{sp}}(c)$ is the expected SP price given $\bar{Z}$ and issuer FE

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- **Note**: $V_i^{\text{sp}}(c)$ is the expected SP price given $\bar{Z}$ and issuer FE
- Identification of $\mu_{\text{pay-up}}$: Exploit discontinuities in the value of customization across loan size cutoffs (l_k)

Wholesale pass-through regression

VARIABLES	(1) Wholesale pay-up	(2) Security pay-up	(3) Wholesale price	(4) Wholesale price
Security pay-up			0.39*** (0.019)	0.44*** (0.073)
$1(L_i > l_k)$	-0.083*** (0.020)	-0.19*** (0.023)		
TBA price			0.86*** (0.030)	0.88*** (0.035)
Observations	12,695	12,695	12,695	12,695
R-squared	0.754	0.833	0.927	0.791
Estimator	OLS	OLS	OLS	IV

Sample restrictions: Loans sold in OB Auctions (2018-2019), within 5000 of SP size cutoff.

Fixed effects: Auction date x High/Low Coupon and Loan size bin x High/Low Coupon.

Additional controls: Borrower characteristics.

- **What about retail P-T?** Zero to closing costs AND rates.

► Retail tables

► Balance tests

Counterfactual: What is the effect of eliminating the adverse selection of multi-issuer pools?

- **Assumptions:** (1) Competitive secondary market, (2) Constant allocation of loans and interest rates, (3) Linear pass-through, (4) Constant securitization cost
- Counterfactual multi-issuer pool prices:

$$\tilde{V}^{tba}(c) = \frac{1}{Q_c} \left[V^{tba}(c) q_{M,c} + \sum_j q_{j,c} V_j^{sp}(c) \right]$$

- Effects of eliminating adverse-selection:
 - $\Delta V^{tba} = \tilde{V}^{tba}(c) - V^{tba}(c) > 0$
 - $\Delta \text{Pay-up} = 0 - \left(V_j^{sp}(c) - V^{tba}(c) \right) < 0$ for SP loans, and 0 for TBA loans
- Change in borrower's upfront price and profits:
 - Calculated using $\Delta \text{Pay-up} / \Delta V^{tba}$ AND pass-through rates $\mu_{\text{pay-up}}^w$ and $\mu_{\text{pay-up}}^r (=0)$
 - Additional details [▶ here](#)

Distribution of retail price changes: eliminating customization benefits small borrowers

Loan Size	$\Delta P_i = P_i^{\text{cf}} - P_i^{\text{obs}}$			
	mean	sd	p10	p90
85000	0.107	0.112	0.001	0.265
110000	0.092	0.106	-0.001	0.224
125000	0.083	0.102	-0.001	0.202
...				
250000	0.053	0.085	-0.004	0.147
325000	0.048	0.081	-0.004	0.139
1000000	0.035	0.069	-0.004	0.112
Total	0.061	0.091	-0.004	0.155

Distribution of issuer profit changes after the elimination of customization

	Issuer size groups			
	(1)	(2)	(3)	(4)
	1-10	11-50	51-100	100+
Change in issuer profits (bps)	-0.08	-0.06	-0.06	0.06
	(0.11)	(0.12)	(0.23)	(0.20)
Change in issuer profits per loan	-172.12	-122.12	-72.04	164.05
	(259.99)	(252.80)	(429.49)	(390.88)
Fraction custom securities	0.27	0.25	0.29	0.08
	(0.15)	(0.20)	(0.24)	(0.19)
Fraction vertical integration	0.45	0.61	0.82	0.84
	(0.37)	(0.39)	(0.31)	(0.32)
Issuer size (/1000)	126.24	27.67	4.89	0.51
	(73.62)	(16.21)	(1.83)	(0.57)

Standard-deviation in parenthesis.

Policy implications

- **MBS market design:** regulators have encouraged larger/more liquid TBA pools
 - Example: *Uniform MBS* reform (2019) merged Fannie/Freddie pools into a single common pool
 - **Our results:** expanding TBA pools has a **pro-competitive effect** $\Rightarrow$ smaller issuers compete more effectively with large customizers
- **Nonbank fragility:** Fed is concerned about the importance and stability of nonbanks
 - **Our results:** large incumbent customization squeezes margins for smaller institutions
 - Small nonbanks rely on TBA market $\Rightarrow$ **depressed** by adverse selection from customization

We document the mechanism $\Rightarrow$ formal policy evaluation requires a structural model of equilibrium responses along the supply chain

APPENDIX

Correlation between issuer average duration and firm characteristics

VARIABLES	(1) Survival FE
Monthly volume (log, standardized)	-0.334*** (0.0684)
Retail origination FE (standardized)	0.205** (0.0804)
Recapture rate FE (standardized)	-0.313*** (0.0784)
1(Zero recapture)	-0.617*** (0.214)
Constant	1.611*** (0.219)
Observations	257
R-squared	0.213

SE clustered by Issuer

Decomposition of the variance in spec-pool pay-ups within segments

	Variance decomposition	
	(1)	(2)
	SD	%
Pay-up deviations	0.81	1.00
Borrower characteristics	0.32	0.15
Issuer	0.30	0.13
Cov: Charact. and Issuer	-0.01	-0.02
Residual	0.69	0.73
Observations	62626	62626

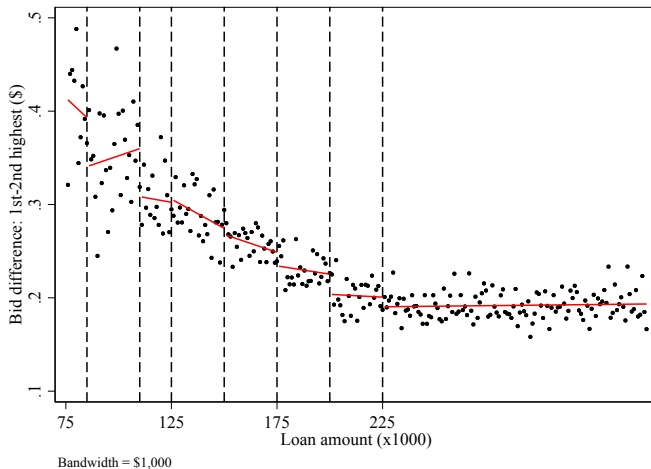
Dep. variable: Pay-up deviation within week and segment (date, size and coupon).

Takeaways:

- Issuer heterogeneity and loan characteristics are equally important
- But important residual price dispersion: Uncertainty in resale values

► Return

Heterogeneity in WTP: bids are more dispersed in auctions for small loans



Sample: Optimal-Blue wholesale auctions from 2018–2019.

[▶ Return](#)

Balance tests: loan and borrower characteristics

	85k	110k	125k	150k	175k
Income	-0.0783 (0.118)	-0.246 (0.115)	0.0959 (0.103)	-0.00836 (0.0744)	-0.0473 (0.0659)
FICO	3.633 (2.548)	2.150 (2.639)	1.266 (1.776)	0.850 (1.385)	-0.592 (1.343)
LTV	0.0389 (0.430)	0.343 (0.448)	0.165 (0.285)	-0.456 (0.265)	-0.231 (0.196)
Note Rate	-0.00112 (0.0353)	0.00502 (0.0326)	0.0390 (0.0279)	0.00110 (0.0230)	0.00712 (0.0204)
1(FHA Loan)	-0.0254 (0.0218)	-0.0145 (0.0204)	0.00820 (0.0207)	-0.000974 (0.0129)	-0.0113 (0.0111)

Retail pass-through regression

VARIABLES	(1) Retail price	(2) Wholesale pay-up	(3) Retail price	(4) Retail price
Wholesale pay-up			0.14*** (0.0064)	-0.037 (0.048)
$1(L_i > l_k)$	0.0084 (0.0091)	-0.12*** (0.020)		
TBA price			0.34*** (0.0080)	0.32*** (0.013)
Observations	19,219	19,219	19,219	19,599
R-squared	0.864	0.553	0.963	0.647
Estimator	OLS	OLS	OLS	IV

Sample restrictions: Loans sold in OB Auctions (2018-2019), within 5000 of SP size cutoff.

Fixed effects: Auction date x High/Low Coupon and Loan size bin x High/Low Coupon.

Additional controls: Borrower characteristics.

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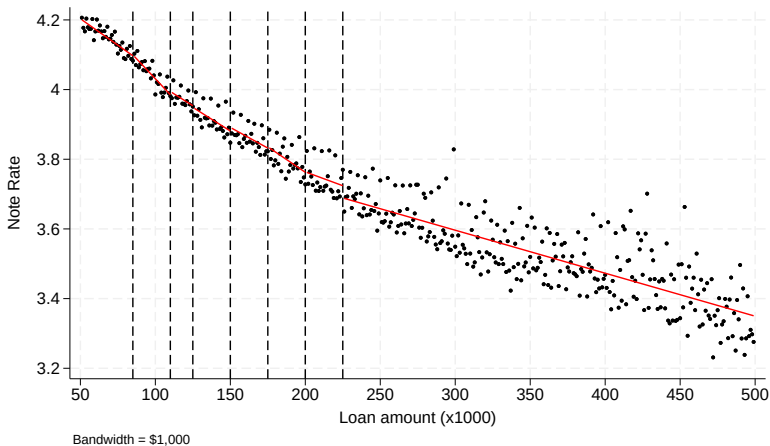
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Interest-rate pass-through: no discontinuity across loan size cutoffs



Counterfactual: What is the effect of eliminating the adverse selection of multi-issuer pools?

- Change in Issuers' profits = Δ (Resale value - Acquisition price)

$$\Delta \Pi_i^{\text{Issuer}} = \begin{cases} (1 - \mu_{\text{pay-up}}^w) \Delta \text{Pay-up} + (1 - \mu_{\text{tba}}^w) \Delta V^{\text{tba}}(c) & \text{Corr.} \\ (1 - \mu_{\text{pay-up}}^r) \Delta \text{Pay-up} + (1 - \mu_{\text{tba}}^r) \Delta V^{\text{tba}}(c) & \text{Retail} \end{cases}$$

Note: $\mu_{\text{pay-up}}^w > \mu_{\text{pay-up}}^r = 0$.

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Note: $\mu_{\text{pay-up}}^w > \mu_{\text{pay-up}}^r = 0$.

- Change in borrower's upfront price:

$$\Delta P_i = \begin{cases} \mu_{\text{pay-up}}^r \mu_{\text{pay-up}}^w \Delta \text{Pay-up} + \mu_{\text{tba}}^r \mu_{\text{tba}}^w \Delta V_{\text{tba}}(c) & \text{Corr.} \\ \mu_{\text{pay-up}}^r \Delta \text{Pay-up} + \mu_{\text{tba}}^r \Delta V_{\text{tba}}(c) & \text{Retail} \end{cases}$$

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